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MAST20030: Differential Equations
Tutorial: Week 11

Question 1

In this question, we're going to solve the 1D wave equation with an initial condition that represents a plucked string. The catch is that we'll pluck it at different places and see what the resulting differences are. Assume:

- The length of the string is 1.
- The string is fixed at both ends.
- Initially, the string is not moving but is plucked with an amplitude A at a position ℓ units along the string ($0 < \ell < 1$).

In other words, we model the initial configuration of the string as two line segments, one joining $(0, 0)$ to (ℓ, A) and the other joining (ℓ, A) to $(1, 0)$.

(a) Write down the complete set of data for this initial-boundary value problem. One of your initial conditions should involve a piecewise-continuous function depending on A and ℓ.

(b) Use separation of variables to derive the following (formal) solution to this problem:

$$u(x,t) = \sum_{n=1}^{\infty} b_n \cos(\pi n c t) \sin(\pi n x) \quad \text{with} \quad b_n = \frac{2A}{\pi^2 n^2} \frac{\sin(\pi n \ell)}{\ell(1-\ell)}.$$

The angular frequencies of the string (the *harmonics*) have the form $\omega_n = \pi c n$, $n = 1, 2, 3, \ldots$ The sound we perceive is ω_1, but the quality of the sound we perceive (its *timbre*) depends on the ratios $|b_n/b_1|$.

(c) Simplify this solution for $\ell = \frac{1}{2}$ and tabulate the first six of the ratios $|b_n/b_1|$.

(d) Repeat the previous tabulation for $\ell = \frac{1}{3}$. How does the choice of ℓ affect the $|b_n/b_1|$ and which harmonics actually appear in the solution?

(e) What happens to the solution and the ratios as $\ell \to 0^+$?

In terms of timbre, plucking a string near its middle results in a relatively empty sound, plucking further down gives a richer, more complex, sound, and plucking very close to its ends results in a harsher metallic sound.